learning algorithm, there is a dataset where it is very accurate and another dataset where it is very poor. This is called the *No Free Lunch Theorem*.[1]

5.1.4 Other factors

Classification algorithms can be compared based not only on error rates but also on several other criteria like the following:

- risks when errors are generalized using loss functions
- training time and space complexity,
- testing time and space complexity,
- interpretability, namely, whether the method allows knowledge extraction which can be checked and validated by experts, and
- easy programmability.

5.2 Cross-validation

To test the performance of a classifier, we need to have a number of training/validation set pairs from a dataset X. To get them, if the sample X is large enough, we can randomly divide it then divide each part randomly into two and use one half for training and the other half for validation. Unfortunately, datasets are never large enough to do this. So, we use the same data split differently; this is called *cross-validation*.

Cross-validation is a technique to evaluate predictive models by partitioning the original sample into a training set to train the model, and a test set to evaluate it.

The *holdout method* is the simplest kind of cross validation. The data set is separated into two sets, called the training set and the testing set. The algorithm fits a function using the training set only. Then the function is used to predict the output values for the data in the testing set (it has never seen these output values before). The errors it makes are used to evaluate the model.

5.3 K-fold cross-validation

In K-fold cross-validation, the dataset X is divided randomly into K equal-sized parts, X_i, $i = 1, \ldots, K$. To generate each pair, we keep one of the K parts out as the validation set V_i, and combine the remaining $K - 1$ parts to form the training set T_i. Doing this K times, each time leaving out another one of the K parts out, we get K pairs (V_i, T_i):

$$
\begin{aligned}
V_1 &= X_1, \quad T_1 = X_2 \cup X_3 \cup \ldots \cup X_K \\
V_2 &= X_2, \quad T_2 = X_1 \cup X_3 \cup \ldots \cup X_K \\
&\ldots \\
V_K &= X_K, \quad T_K = X_1 \cup X_2 \cup \ldots \cup X_{K-1}
\end{aligned}
$$

Remarks

1. There are two problems with this: First, to keep the training set large, we allow validation sets that are small. Second, the training sets overlap considerably, namely, any two training sets share $K - 2$ parts.

[1] "We have dubbed the associated results NFL theorems because they demonstrate that if an algorithm performs well on a certain class of problems then it necessarily pays for that with degraded performance on the set of all remaining problems."(David Wolpert and William Macready in [7])

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2. K is typically 10 or 30. As K increases, the percentage of training instances increases and we get more robust estimators, but the validation set becomes smaller. Furthermore, there is the cost of training the classifier K times, which increases as K is increased.

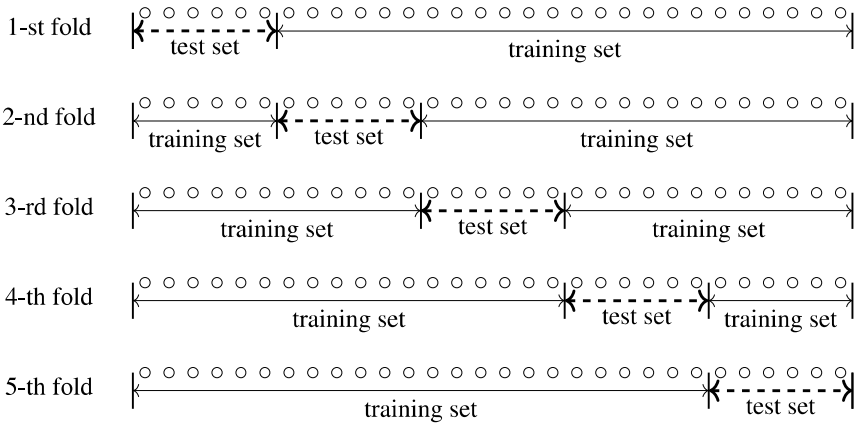


Figure 5.1: One iteration in a 5-fold cross-validation

Leave-one-out cross-validation

An extreme case of K-fold cross-validation is *leave-one-out* where given a dataset of N instances, only one instance is left out as the validation set and training uses the remaining $N - 1$ instances. We then get N separate pairs by leaving out a different instance at each iteration. This is typically used in applications such as medical diagnosis, where labeled data is hard to find.

5.3.1 5×2 cross-validation

In this method, the dataset X is divided into two equal parts $X_1^{(1)}$ and $X_1^{(2)}$. We take as the training set and $X_1^{(2)}$ as the validation set. We then swap the two sets and take $X_1^{(2)}$ as the training set and $X_1^{(1)}$ as the validation set. This is the first fold. the process id repeated four more times to get ten pairs of training sets and validation sets.

$$
\begin{aligned}
T_1 &= X_1^{(1)}, \quad V_1 = X_1^{(2)} \\
T_2 &= X_1^{(2)}, \quad V_2 = X_1^{(1)} \\
T_3 &= X_2^{(1)}, \quad V_3 = X_2^{(2)} \\
T_4 &= X_2^{(2)}, \quad V_4 = X_2^{(1)} \\
&\vdots \\
T_9 &= X_5^{(1)}, \quad V_3 = X_5^{(2)} \\
T_{10} &= X_5^{(2)}, \quad V_{10} = X_5^{(1)}
\end{aligned}
$$

It has been shown that after five folds, the validation error rates become too dependent and do not add new information. On the other hand, if there are fewer than five folds, we get fewer data (fewer than ten) and will not have a large enough sample to fit a distribution and test our hypothesis.

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